

SWAPNANIL BALA

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

M.S., Data Science

Sep 2025 - Dec 2027

Boston, MA

PROJECTS

Lagna Atelier | lagnaatelier.site

- Vedic Astrology Chart Engine
- Built and deployed a TypeScript PWA that computes full Vedic charts client-side, with Google OAuth and a NeonDB backend for saved charts.
- Hand-wrote a sidereal ephemeris on astronomy-engine with IAU-2006 precession, supporting 6 ayanamsha models and 6 house systems so users can pick a tradition and compare results.
- Layered derived calculations on top: 23 divisional charts, collapsible Vimshottari dasha timelines, and 51 yoga detections.
- Defined yogas as data through a Zod-validated rule DSL rather than hardcoded logic, and ran a Monte Carlo harness over simulated charts to measure each rule's empirical rarity.
- Shipped AI chart guidance (career, relationships, family, inheritance) and photo-based palm reading on Claude Opus 5 vision with MediaPipe hand landmarking, behind rate-limited edge functions.
- Optimized delivery with service workers, web workers, and LRU/TTL caching; localized to 6 languages; backed by 561 tests.

Robust Health | app.robusthealth.in

- AI Fitness & Wellness Platform
- Shipped a multi-role Next.js PWA with member and trainer portals - members get generated plans; trainers author and assign their own, and track client adherence over adjustable time windows; localized to English, Spanish, and Portuguese.
- Generated personalized weekly training, nutrition, and recovery plans through the Anthropic API, then revised each plan the following week against what the user actually logged.
- Built a powerlifting module with RPE logging, auto-calculated working weight for the next set, a dedicated analytics dashboard, and flags for clients who stop logging.
- Modeled 15 training programs and 160+ exercises as structured data (sets, reps, rest, RPE-based progression) and computed calorie targets with the Mifflin-St Jeor equation.
- Implemented auth with Supabase Row-Level Security, OAuth 2.0 (Google, GitHub, Discord), and one-time email codes; cut Lighthouse load time ~50% on web and ~67% on mobile via next/image compression and code splitting.

EXPERIENCE

P2G Mobility Tech Private Limited (Pointo) - EV-Financing Startup

Nov 2024 - Jan 2025

Program Analyst Intern, Technology

Kolkata, India

- Cleaned and consolidated operational Excel datasets feeding internal dashboards - resolving missing values, recovering lost source files, and standardizing records for weekly reporting.
- Wrote SQL against the production database and built recurring operational reports for the Project Lead.
- Supported field customer acquisition by pitching the paytoown financing program to erickshaw operators and explaining lithiumion upgrades over legacy leadacid batteries, helping attract new operators to the program

SKILLS

- **Languages:** TypeScript, Python, JavaScript, SQL
- **ML & AI:** PyTorch, Scikit-Learn, XGBoost, transformer fine-tuning (DistilBERT, RoBERTa), NLP, SHAP explainability, Anthropic Claude API (text + vision), MediaPipe, multimodal inference
- **Data & Analytics:** Pandas, NumPy, EDA, hypothesis testing (t-test, chi-square, ANOVA), Matplotlib, Seaborn
- **Web & APIs:** React, Next.js, Node.js, REST API design, edge functions, PWAs (service workers, web workers), OAuth 2.0, i18n, Lighthouse optimization
- **Databases & DevOps:** PostgreSQL (NeonDB, Supabase), MySQL, schema design and normalization, stored procedures, Row-Level Security, Zod validation, Vitest/Jest, Git, Docker, Vercel

CERTIFICATIONS

Meta Database Engineer Professional Certificate: Coursera, Jan 2025

Jan 2025

- nine-course program covering relational design and normalization, advanced MySQL (stored procedures, triggers, views, query optimization), Python database clients, data modeling and warehousing, and Git, closing with a capstone that built and queried a production-style schema end to end.

Google Advanced Data Analytics Professional Certificate

- TikTok claims classification: NLP pipeline flagging policy-violating content at ~99.5% recall (5 misclassifications across 3,817 test cases). Waze churn prediction: EDA and hypothesis testing (t-test, chi-square, ANOVA); tuned logistic regression, Random Forest, and XGBoost with GridSearchCV.

ACHIEVEMENTS

- **ABRSM Grade 4 Piano:** Distinction, November 2024